

CaSKG Pipeline: Skill Graph Construction and Task-Aware Retrieval

Unreliable Skill-Relation Bottleneck

- Semantic similarity does not imply operational dependency
- Co-occurrence hides source-to-target direction
- Uniform edge weights propagate unreliable relevance
- Exhaustive counterfactual validation is costly

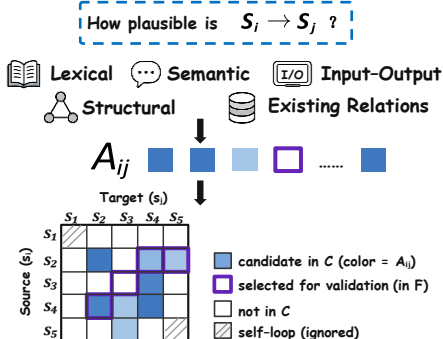
Graph-Construction Inputs

- Lexical similarity from skill descriptions
- Semantic similarity from skill embeddings
- Input-output compatibility
- Structural signals from workflow roles and positions
- Existing relation records

problem driven

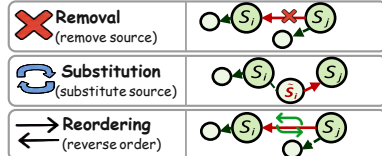
evidence inputs

1 Candidate Skill Graph Induction



2 Counterfactual Edge Probing

Remove / Substitute / Reorder →
Necessity / Specificity / Order Dependence



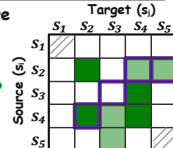
Counterfactual support score

$$P_{rem}(s_i, s_j) = (\emptyset, s_j),$$

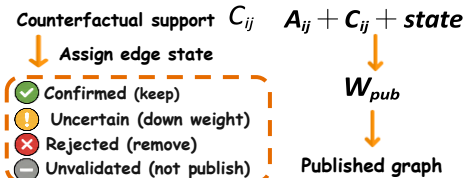
$$P_{sub}(s_i, s_j) = (\hat{s}_i, s_j),$$

$$P_{ord}(s_i, s_j) = (s_j, s_i)$$

$$e_{ij}^{(m)}$$



3 Edge States & Publication



4 Task-Conditioned Skill Retrieval

